WinNovation

EXECUTIVE MASTER PROGRAM 2026-27

PREMIER TECHNOLOGY & SKILL EDUCATION

DATA STRUCTURES & ALGORITHMS (DSA)

Master Algorithmic Problem Solving, Linear & Non-Linear Structures, Advanced Trees, Dynamic Programming, Graph Theory & FAANG Technical Interview Defense through 70% live coding sprints and production software architecture.



3 MONTHS	70 : 30	250+	100%
12 WEEKS INTENSIVE	LIVE CODING : THEORY	LEETCODE PROBLEMS	PLACEMENT SUPPORT

Complexity & Big-O • Linear Structures • Trees, BST & Heaps • Graphs & Dynamic Programming • FAANG Interview Prep

WINNOVATION EDUCATION PVT. LTD.
Building Careers Since 2017 | Greater Noida, Delhi NCR

Hotline: **+91 92112 28030** | Email: **admissions@winnovation.org**
Web: **www.winnovation.org** | Campus: **1157, i-Thum Galleria, Alpha 2**

Accelerate Your Career with Winnovation

Bridging the divide between theoretical textbook pseudocode and high-throughput production software engineering.



*"At Winnovation, our purpose is driven by the **future of our youth**. In today's competitive global technology ecosystem, mastering **Data Structures & Algorithms (DSA)** is the foundational bedrock of world-class software engineering. We established Winnovation to bridge the divide between theoretical textbook pseudocode and real-world high-throughput software architecture—ensuring our students write mathematically optimized, production-ready code with **optimal time and space complexity** to ace elite product engineering interviews."*

Sarbarinder Singh

Director, Winnovation Education Pvt. Ltd.

"Win With Winnovation" | Recognized Skill Incubator

NEP 2020 & Competency-Based Education (CBE) Blueprint

Winnovation curricula strictly align with India's **National Education Policy (NEP 2020)** and **CBSE Vocational Skill Education** paradigms. We replace rote memorization with experiential problem solving using Bloom's Taxonomy.

70% Live Coding Labs

Rigorous daily problem solving in C++, Java, and Python on LeetCode and Codeforces with formal asymptotic time-space benchmarking.

Mathematical & Algorithmic Rigor

Formal asymptotic analysis, amortized bounds, recurrence relations (Master Theorem), loop invariants, and provable correctness.

Outcome-Driven Portfolios

Every module produces verifiable public software systems (custom memory allocators, graph navigation engines, priority task schedulers).

CBSE Vocational Assessment Scheme (100 Marks Blueprint)

Assessment Component	Weightage	Evaluation Criteria	Frequency
Continuous Formative Coding Sprints	30 Marks	Weekly timed LeetCode-style problem sets, code reviews, complexity proofs	Weekly Milestones (12 Sprints)
Mid-Term Algorithmic Hackathon	20 Marks	Live 4-hour competitive programming sprint on trees, recursion & heaps	End of Month 1 & Month 2
Comprehensive Production Capstone	20 Marks	End-to-end engineered software engine (Graph Routing, Cache, B-Tree Indexer)	Final Capstone Showcase
Certified Code Repository & Journal	15 Marks	GitHub repository with clean modular code, unit tests & benchmark metrics	Terminal Evaluation
Technical Whiteboard & Viva Voce	15 Marks	Oral defense solving unannounced algorithmic challenges before SDE panel	Final SDE Jury Review

Surging Industry Demand for Software Problem Solvers

Powering high-performance backend systems, distributed architectures, and elite product engineering across global tech enterprises.

88%

Product Companies Prioritize DSA

Global tech leaders (Google, Amazon, Microsoft, Uber) and tier-1 Indian unicorns use algorithmic problem solving as their primary hiring filter.

3.5×

Higher Compensation Multiplier

Engineers who master asymptotic complexity, dynamic programming, and systems design command 3.5× higher compensation than standard framework developers.

₹18.5 LPA

Average Product SDE Starting CTC

Leading product companies and tech unicorns offer substantial starting packages with aggressive stock incentives and joining bonuses.

Industry Salary Potential in India (AmbitionBox & Levels.fyi Benchmarks)

Entry-Level (0 – 2 Years)

₹8 – ₹16 LPA

Target Roles: Junior Software Development Engineer (SDE-1), Associate Systems Engineer, Graduate Tech Trainee.

Mid-Level (3 – 6 Years)

₹16 – ₹32 LPA

Target Roles: SDE-II, Backend Systems Engineer, Algorithm Optimization Lead, Distributed Systems Engineer.

Senior-Level (7+ Years)

₹32+ – ₹60+ LPA

Target Roles: Senior SDE, Principal Software Architect, Staff Engineer, VP of Technology & Infrastructure.

Who Should Pursue This Program?

College Students & CS Undergraduates

B.Tech CSE/IT, BCA, MCA, and B.Sc CS students preparing for campus placements, competitive programming, and off-campus FAANG drives.

Service-to-Product Company Switchers

Software professionals working in IT service firms looking to break into high-paying tier-1 product companies and tech unicorns.

Non-CS Engineers & Career Changers

STEM graduates from Mechanical, Electrical, or Civil disciplines wanting to transition into mainstream software engineering with strong fundamentals.

Competitive Programming Aspirants

Coders targeting 1800+ ratings on LeetCode and Codeforces, aiming for ICPC regional algorithmic competitions and hackathon victories.

THE ECONOMIC TIMES: "Data Structures, Algorithms & System Architecture remain the single most evaluated skill set across India's top product companies."

GARTNER: "Foundational computational problem solving outweighs familiarity with ephemeral software frameworks by 4 to 1."

Roadmap to Get Your Dream Tech Job

A step-by-step competency ladder transforming aspiring coders into elite algorithmic problem solvers and product-ready software engineers.

1

Asymptotic Analysis & Algorithmic Foundations

Weeks 1 – 2

Master Big-O, Big-Omega, Big-Theta notations, space-time trade-offs, call stack mechanics, recursion tree analysis, and the Master Theorem for divide-and-conquer recurrences.

2

Linear Data Structures & Two-Pointer Patterns

Weeks 3 – 4

Implement dynamic arrays, 2D matrices, prefix sums, sliding window paradigms, and singly/doubly linked lists with Floyd's cycle detection and reverse in K-groups.

3

Stacks, Queues & Hierarchical Tree Systems

Weeks 5 – 6

Build monotonic stacks, circular queues, hash maps with collision resolution, binary search trees (BST), balanced AVL trees, and min/max priority heaps.

4

Advanced Graph Theory & Network Algorithms

Weeks 7 – 8

Master BFS, DFS, cycle detection, topological sort (Kahn's), Dijkstra's shortest path, Bellman-Ford, Prim's and Kruskal's minimum spanning trees, and Disjoint Set Union (DSU).

5

Dynamic Programming & Greedy Optimization

Weeks 9 – 10

Solve complex overlapping subproblems using memoization and tabulation, 0/1 knapsack, longest common subsequence (LCS), longest increasing subsequence (LIS), and greedy interval scheduling.

6

System Design, Whiteboarding & FAANG Interview Defense

Weeks 11 – 12

Simulate 1-on-1 technical whiteboard rounds, behavioral STAR responses, clean production Git architecture, and live algorithmic defense before senior software engineering panels.

The CAISEN Immersive Learning Cycle

1. STUDY
Live masterclasses & 1:1 expert mentoring with 24×7 recordings.

2. ATTEMPT
Daily timed problem sets & algorithmic proofs on LeetCode.

3. MAKE
Implement production data structures & memory engines from scratch.

4. REVISE
Code reviews, asymptotic optimizations & unit testing benchmarks.

5. TEST
Mock whiteboard rounds before senior engineering leaders & FAANG juries.

Curriculum Blueprint: Month 1 & 2

Algorithmic foundations, mathematical complexity, linear structures, and hierarchical tree systems.

Month 1: Asymptotic Complexity & Algorithmic Foundations

Weeks 1 – 4 | 40 Contact Hours

Core Theoretical & Applied Topics

- Complexity Analysis: Big-O, Big-Omega, Big-Theta notations, worst, average, and amortized time complexity
- Recursion Mechanics: Call stack execution, base conditions, recursion trees, tail recursion optimization
- Array & Matrix Algorithms: 1D/2D arrays, prefix sums, Kadane's algorithm, sliding window, two-pointer techniques
- Searching & Sorting: Binary search on answer space, Merge Sort, Quick Sort, QuickSelect, Inversion count
- Backtracking Foundations: Permutations, combinations, subsets generation, N-Queens problem, Sudoku solver
- Mathematical Algorithms: GCD, Sieve of Eratosthenes, fast modular exponentiation, bitwise manipulation

Guided Laboratory Practical & Takeaways

- **Lab 1:** Big-O Complexity Benchmarking & Memory Profiling Suite in C++/Java
- **Lab 2:** Multi-Dimensional Array Manipulation & Subarray Optimization Engine
- **Lab 3:** Recursive Backtracking Engine with Pruning for N-Queens & Combinatorial Search
- **Competency:** Algorithmic time-space analysis, divide-and-conquer strategies & backtracking

Milestone Capstone 1: High-Performance In-Memory Search Engine with O(log N) Binary Retrieval

Engineered in C++/Java to index and retrieve 100,000+ records using custom binary search on answer space, QuickSelect median finding, and memory-efficient dynamic partitioning.

Month 2: Core Linear & Hierarchical Data Structures

Weeks 5 – 8 | 40 Contact Hours

Core Theoretical & Applied Topics

- Linked Structures: Singly, doubly, and circular linked lists, Floyd's cycle detection, reverse in K-groups
- Stacks & Monotonic Patterns: Stack implementation via arrays/nodes, Next Greater Element, Min-Stack, trapping rain water
- Queues & Deques: Circular queues, double-ended queues (deque), sliding window maximum, monotonic queues
- Hash Tables & Collision: Hash functions, open addressing (linear/quadratic probing), chaining, rolling hash (Rabin-Karp)
- Binary Trees: Preorder, inorder, postorder, level-order traversals, lowest common ancestor (LCA), tree diameter
- Binary Search Trees (BST): BST search, insertion, deletion, inorder predecessor/successor, balanced BST concepts

Guided Laboratory Practical & Takeaways

- **Lab 4:** Memory-Efficient Doubly Linked List with O(1) LRU Cache Eviction Node Linking
- **Lab 5:** Monotonic Stack & Queue Engine for Real-Time Streaming Data Window Analysis
- **Lab 6:** Hash Map Engine with Custom Dynamic Resizing & Prime Bucket Collision Handling
- **Competency:** Linear memory management, pointer manipulation, hashing architectures & tree traversals

Milestone Capstone 2: Smart Student Record Management System with B-Tree Indexing & Hash Tables

A multi-threaded data management engine delivering O(1) student record lookups via custom hash maps and O(log N) range queries using self-balancing B-Tree structures.

Curriculum Blueprint: Month 3 & Algorithms

Advanced graph theory, dynamic programming, priority queues, and FAANG technical interview preparation.

Month 3: Advanced Trees, Graphs, Dynamic Programming & SDE Interviews

Weeks 9 – 12 | 40 Contact Hours

Advanced Trees, Heaps & Graph Theory

- Balanced Trees & Heaps: AVL trees, Red-Black principles, Binary Min/Max Heaps, HeapSort, Top-K streaming
- Graph Traversal: Adjacency lists/matrices, BFS, DFS, bipartite verification, cycle detection
- Topological Sorting: Kahn's algorithm, DAG scheduling, strongly connected components (Kosaraju)
- Shortest Paths: Dijkstra's algorithm with priority queues, Bellman-Ford, Floyd-Warshall all-pairs shortest paths
- Minimum Spanning Trees: Prim's and Kruskal's algorithms, Disjoint Set Union (DSU) with path compression
- Memory Layout: Cache-conscious node structures, pointer indirection vs cache locality profiling

Dynamic Programming & SDE Interview Mastery

- DP Paradigms: Memoization vs Tabulation, state definitions, base cases, transition functions
- Classical DP Patterns: 0/1 Knapsack, Unbounded Knapsack, Longest Common Subsequence (LCS), LIS
- Multi-Dimensional DP: Grid paths, matrix chain multiplication, DP on trees and subtrees
- Greedy Paradigms: Interval scheduling, Huffman coding, fractional knapsack, activity selection
- FAANG Technical Sprints: Daily LeetCode Hard problem dissection, live whiteboard coding simulations
- SDE System Defense: Behavioral STAR framework, code clarity, unit test writing, and complexity justification

Milestone Capstone 3: Autonomous Graph-Based GPS Navigation Engine & Distributed Priority Task Scheduler

An enterprise-grade routing and scheduling system implementing Dijkstra's algorithm and A* heuristics on road network graphs, combined with a min-heap task dispatcher executing concurrent workloads.

Industry Standard Algorithmic Tools & Engineering Stack

Programming Languages & Libraries

- C++ (C++20, STL: vectors, maps, heaps)
- Java (Java 17+, Collections Framework)
- Python (collections, deque, heapq, bisect)
- Low-Level Profiling (Valgrind, GDB Debugger)

Competitive Platforms & Contests

- LeetCode (250+ Curated SDE Problem Sets)
- Codeforces & CodeChef (Div 2/3 Contests)
- HackerRank & GeeksforGeeks Benchmarks
- GitHub Portfolio & Open Source Contributions

Developer Tools & Whiteboarding

- VS Code, CLion & IntelliJ IDEA
- Git & GitHub Version Control
- Pramp & Interviewing.io Mock Platforms
- Excalidraw & Draw.io Whiteboard Architecture

Certification Alignment: Fully prepares candidates for global software engineering benchmarks, EC-Council Certified Secure Software Lifecycle, Google Cloud Professional Developer tracks, and AWS Certified Developer Associate credentials.

FAANG & Tier-1 Ready

Industry-Ready Project Portfolio

Production-grade algorithmic engines and distributed software architectures designed for maximum scalability and speed.

Project 1: Distributed High-Concurrency In-Memory Key-Value Store & LSM-Tree Engine

Modern C++20 + LSM-Trees + SkipLists & Bloom Filters + WAL

Business Problem: High-throughput distributed storage engines require sub-millisecond key-value lookups, high-concurrency writes, and persistent disk serialization without suffering write amplification or GC pauses.

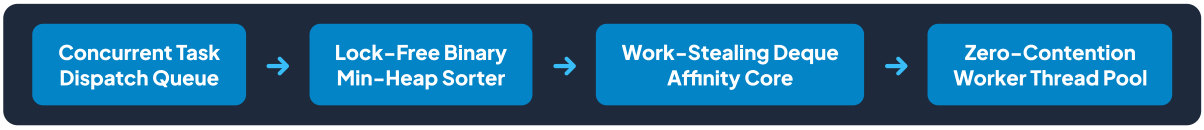


Deliverable: Production C++ storage engine delivering 150,000+ QPS, O(1) probabilistic Bloom filter membership checks, cache-conscious SkipList indexing, zero-copy socket I/O, and Valgrind memory leak certification.

Project 2: Real-Time Multi-Threaded Distributed Task Scheduler & Priority Work-Stealing Pool

Java 21 / Modern C++ + Lock-Free Deques + Atomic CAS + Heaps

Business Problem: Cloud computing hypervisors and distributed task queues require zero-lock contention, optimal CPU core affinity, and microsecond-level priority job dispatching under 100% thread saturation.



Deliverable: High-throughput work-stealing runtime with Chase-Lev lock-free deque algorithms, microsecond-level task latency, priority aging to prevent starvation, and JMH benchmark profiling.

Project 3: High-Scale Geospatial Graph Routing Engine & Multi-Modal Detour Topology System

C++20 STL + Contraction Hierarchies + Bidirectional A* + DSU

Business Problem: Autonomous navigation networks require sub-10ms point-to-point shortest path queries across 10,000,000+ road intersection vertices with dynamic live traffic weight adjustments and turn penalties.



Deliverable: Scalable graph computation engine achieving sub-5ms path finding across multi-million node graphs via contraction hierarchies, memory-compact adjacency arrays, and Kruskal's MST for utility networks.

Industry-Recognized Certifications

4 official credentials validating your software engineering, algorithmic problem solving, and cloud development expertise to global recruiters.



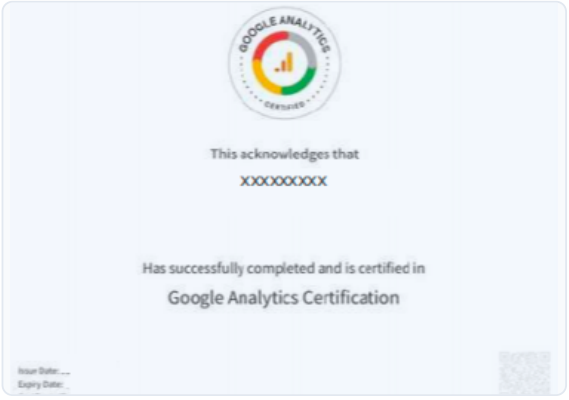
Winnovation Master Certificate in DSA

Awarded on successful completion of 12 weeks of training, 250+ solved algorithmic problems, and jury defense of all 3 milestone capstones. Features verified QR code.



Executive Software Developer Credential

ANSI-accredited international certification validating secure software development lifecycles, memory safety standards, and robust algorithmic coding practices.



Google Cloud Associate Developer Credential

Industry gold standard credential verifying proficiency in scalable cloud applications, containerized services, microservices design, and optimal cloud database indexing.



Cloud Infrastructure & Systems Practitioner

Official Amazon Web Services credential validating foundational knowledge of cloud data stores, low-latency queues, serverless compute, and distributed architectures.

Curriculum Endorsed & Co-Accredited By Global Learning Standards

Google Cloud Recognized • AWS Academy Partner • NASSCOM FutureSkills Prime • EC-Council Accredited • IEEE Standards Aligned

Learn from Industry Experts

Seasoned software architects, competitive programming mentors, and algorithms leads guiding you through every problem set, code review, and mock interview.



Abhishek Mishra

8+ Yrs Exp

Senior Software Architect & Algorithms Lead

Specializes in high-concurrency C++ architectures, STL optimization, advanced graph traversals, and asymptotic performance profiling.

C++ / STL | Advanced Graphs | Asymptotic Profiling



Surekha Kanwar

10+ Yrs Exp

Senior Systems Engineer & Algorithmic Lead

Expert in dynamic programming optimizations, recurrence relations, distributed data structures, and mathematical proofs of correctness.

Dynamic Programming | Java Collections | Complexity Theory



Saurabh Dwivedi

5+ Yrs Exp

Senior Cloud Backend & Systems Specialist

Designs distributed hash tables, priority task queues, and low-latency data structures for high-throughput cloud microservices.

Priority Queues | Distributed Trees | Cloud Systems



Nisha Pal

5+ Yrs Exp

Senior Low-Level Code & Visualizer Specialist

Specializes in pointer manipulation, custom memory allocators, memory leak profiling, and interactive data structure visualizers.

Memory Allocators | Pointer Mechanics | C++ Systems



Arun Kumar

5+ Yrs Exp

Senior Systems & Bit Manipulation Engineer

Focuses on cache-conscious algorithmic paradigms, bit manipulation tricks, fast mathematical algorithms, and Python data structures.

Bit Manipulation | Cache Locality | Python DSA



Priyanka Saini

7+ Yrs Exp

Competitive Programming & Problem Solving Lead

Guides students through LeetCode Hard problem patterns, recursive backtracking with pruning, and advanced tree balancing mechanics.

LeetCode Hard | Tree Balancing | Backtracking



Pavneet Kaur

5+ Yrs Exp

Senior SDE Technical Interview & Career Coach

Career strategist specializing in FAANG whiteboarding defense, behavioral STAR frameworks, and elite technical communication.

Whiteboard Defense | FAANG Strategy | SDE Careers



Diksha Sachan

8+ Yrs Exp

Head of Technical Curriculum & Problem Solving

Directs daily competitive coding sprints, greedy strategy design, interval scheduling, and sliding window optimization patterns.

Greedy Algorithms | Sliding Window | Problem Sets



Prince Pal

5+ Yrs Exp

Enterprise Code Governance & Security Lead

Ensures enterprise code quality standards, memory safety verification via Valgrind, and secure software development lifecycle practices.

Valgrind Profiling | Memory Safety | Secure Coding

1:1 Dedicated SDE Mentorship & Live Whiteboard Reviews

Weekly personalized office hours, live algorithmic code reviews, and mock FAANG whiteboard interview sessions tailored to your target tech role.

50+ Hours Live Mentorship

Student Success Stories

Hear how Winnovation graduates cracked elite product engineering roles and accelerated their tech careers through DSA mastery.



Syed Vernice Kashani

B.Tech (CSE)

Software Development Engineer (SDE-1)

"Mastering DSA at Winnovation transformed my problem-solving ability. Solving 300+ LeetCode problems with mentor feedback gave me the confidence to crack high-pressure whiteboard rounds at a leading tech firm."



Prabjyot Singh

B.Tech (CSE)

Backend Systems Engineer

"Winnovation's emphasis on time-space complexity and production implementations made all the difference. The faculty broke down complex topics like Dynamic Programming and Graphs into crystal-clear mental models."



Rabinder Singh

Govt. Technical Trainer

Senior Technical Problem Solving Trainer

"The structured curriculum and daily coding sprints at Winnovation are second to none. I gained deep mastery over advanced tree hierarchies, heaps, and graph algorithms that elevated my teaching and development career."



Suhaib Muzaffar

B.Tech (CSE)

Algorithmic Software Engineer

"I was intimidated by competitive programming before joining Winnovation. The mentors taught us how to recognize patterns like sliding window, monotonic stacks, and topological sort. I secured an amazing SDE role!"



Lakshay Suryavanshi

RA at NextX Tech

High-Performance Systems Engineer

"Building production capstones like the B-Tree Student Record Manager and Graph Navigation Engine set my resume apart from other candidates. The placement cell prepared me with rigorous mock whiteboard interviews."



Aryan Choudhary

B.Tech (CSE)

Junior SDE / Systems Developer

"The DSA Master Program gave me the exact foundation product companies test for. From recursion trees to multi-threaded task scheduling, the practical training was instrumental in cracking my campus placement."

100% Placement Record: Highest SDE Package: ₹24 LPA | Average Starting CTC: ₹9.5 LPA Verified Alumni Network

Backed by 100+ Hiring Partners

Leading product enterprises, high-growth tech startups, and Indian multinational corporations recruiting Winnovation software engineers.

HONDA

adani Ports and Logistics

SAMSUNG

airtel

LUMINOUS Enjoy Technology

Akasa Air

SOTI

vivo

HEIZEN

SOPHOS

UTKARSH SMALL FINANCE BANK

XONIER TECHNOLOGIES

Data Resolve

DP WORLD

FORTRA

Emcure

we360.ai

ALKEM

Medtronic

arcon

motadata

FORTINET

100+

Active Corporate Recruiters

Product Tech Giants, Unicorns & Global IT MNCs

15+

Interview Drives Annually

Exclusive On-Campus & Direct Hiring Drives

100%

Placement Assistance

GitHub reviews, mock whiteboards & SDE interviews

Holistic Guidance & Admissions

The complete CAISEN support framework powering your journey from aspiring coder to elite software engineer.

CAISEN 10 Core Competencies for Algorithmic & Software Excellence

Algorithmic Logic | Asymptotic Complexity | Linear Structures | Non-Linear Trees | Graph Networks | Dynamic Programming | Memory Profiling | Clean Code Architecture | Competitive Speed | FAANG Whiteboarding

100% SDE Placement Support

Dedicated placement cell conducting exclusive product company hiring rounds until successful career placement.

AI & GitHub Portfolio Building

ATS-compliant tech resume architecture, GitHub repository optimization, and production code audits by senior architects.

Rigorous Mock Whiteboards

Simulated 1-on-1 technical whiteboard rounds, live problem-solving defense, and behavioral STAR interviews before SDE heads.

Lifetime Code Repository Access

Unlimited access to high-definition recorded masterclasses, editorial solutions, and continuously updated LeetCode problem banks.

Personal Tech Mentorship

Individual career mentoring, competitive coding contest strategy, and guidance to overcome algorithmic bottlenecks.

24×7 Coding Community & TAs

Dedicated Discord and WhatsApp peer discussion channels with active teaching assistants resolving algorithmic queries within minutes.

The Winnovation Technology Ecosystem

Data Structures & Algorithms (DSA) | Data Science & AI/ML | Data Analytics & BI | Full Stack Web Development | Digital Marketing & Growth | Cloud Computing & DevOps | Internet of Things (IoT) | Cyber Security & CEH | Robotic Process Automation (RPA)

Ready to Master Data Structures & Algorithms?

Website: www.winnovation.org
Admissions Email: admissions@winnovation.org | abhishek@winnovation.org
Mobile & Helpline: +91 92112 28030 / +91 92112 28031
Campus Headquarters: 1157, i-Thum Galleria, Alpha 2, Greater Noida, Uttar Pradesh - 201308

ADMISSIONS OPEN

Batch Size: 25 Students Max

APPLY TODAY